

Kristoffer Sahlin, PhD

Associate Professor, Department of Mathematics, Stockholm University

SciLifeLab Fellow, Science for Life Laboratory, Solna, Sweden

ksahlin@math.su.se | +46 70 457 6727 | sahlingroup.github.io

Current Appointments

Associate Professor, Department of Mathematics, Stockholm University (2025–present)

SciLifeLab Fellow, Science for Life Laboratory, Solna (2020–present)

Employment

Assistant Professor, Department of Mathematics, Stockholm University (2020–2025)

Postdoctoral Researcher, University of Helsinki (2019) — Mentor Veli Mäkinen

Postdoctoral Researcher, Pennsylvania State University (2015–2019) — Mentor Paul Medvedev

Education and Training

Ph.D. in Computer Science, KTH Royal Institute of Technology (2010–2015)

Algorithms and statistical models for scaffolding contig assemblies and detecting structural variants using read pair data

Advisor Lars Arvestad, Co-advisor Joakim Lundeberg

M.Sc. in Mathematical Statistics, Stockholm University (2008–2010)

Estimating convergence of Markov chain Monte Carlo simulations

Advisor Sebastian Höhna

B.Sc. in Mathematics, Stockholm University (2005–2008)

Splines: A theoretical and computational study

Advisor Hans Rullgård

Research Funding

KAW — DDLAS PhD project grant 2025–2028 (3.25 MSEK, main PI)

Swedish Research Council (VR) — Start-up grant 2021–2024 (4 MSEK, main PI)

SFO Start-up Funding, Stockholm University 2020–2025 (16 MSEK)

Marie Skłodowska-Curie ETN Grant, 2022–2025 (30 MSEK, co-PI)

NBIS Long-term Infrastructure Funding, 2022– (main PI, strobealign development)

KTH Opportunities Fund, 2014 (Travel grant 12,500 SEK)

Visiting Research Experience

University of Helsinki (Finland, 2014) — Scaffolding and gap-filling of assemblies

Pennsylvania State University (USA, 2014) — Structural variation detection

Teaching

Graduate level

Computational Mathematics (7.5 ECTS)

Applied Programming for Life Sciences (7.5 ECTS)

Undergraduate level

Programming Techniques for Mathematicians (7.5 ECTS)

Advising

PhD Students: Dr. Alexander Petri (2020–2025), Ivan Tolstoganov (2022–), Sami Mahmud Ayden (2023–), Nicolas Buchin (2025–)

Master's Students: Benjamin Maier (2022), Natasha Stopa (2019–2020)

Bachelor's Students: Josefine Röhss (2014), Sebastian Lindbom (2022), Florence Hugh (2024), Rebecca Nikkelsen (2025)

Visiting Researchers: Aryan Soltani, Moein Karimi (2022), Thomas Baudeau, Parisa Azisi (2023)

Academic Service

Committees & Boards:

HiTSeq Organizing Committee (2023–); RECOMB Program Committee (2024–); RECOMB-Seq Program Committee (2020–); Editorial Board member of the journal *Genome Biology* (2025–); RECOMB-Seq Co-Chair (2026); Digital Futures Working Group (2022–2025); Campus Solna Science Talks (2023); Swedish Bioinformatics Workshop Advisory Board (2023).

PhD Thesis Committees: Mohammadreza Mohaghegh Neyshabouri (KTH, 2023), Oxana Sachenkova (Stockholm University, 2023), Yue Liu (KTH, 2024), Joshua Daniel Rubin (Technical University of Denmark, 2025, Opponent)

Reviewer: *Nature Methods*, *Nature Computational Science*, *Nature Communications*, *Genome Research*, *Genome Biology*, *Bioinformatics*, *PLOS Computational Biology*, *BMC Bioinformatics*, *GigaScience*, *NAR Genomics and Bioinformatics*, *Ecology and Evolution*, *Communications in Statistics*.

Invited talks

SeqBIM, Rennes, France, 2024 - Efficient sequence similarity searches with strobemers and applications in read mapping (Keynote talk)

ISCBacademy, Online, 2022 - Efficient sequence similarity searches with strobemers and applications to read mapping <https://www.youtube.com/watch?v=DS4tURz1Wio>.

SMRT Leiden 2021, Online, 2021 - uLTRA: Accurate spliced alignment of long RNA sequencing reads